Tutorial 1 Solutions

Optimization for Machine Learning (E0 331)

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1. Let $\Omega \subset \mathbb{R}^n$ be a convex set. Show that the interior $\text{int}(\Omega)$ and the closure $\overline{\Omega}$ are also convex sets.

   Proof. Let $\Omega \subset \mathbb{R}^n$ be convex.

   Convexity of $\text{int}(\Omega)$**:** Fix $a, b \in \text{int}(\Omega)$ and $\lambda \in (0, 1)$. Since $a \in \text{int}(\Omega)$, there exists an open set $V_a \subset \mathbb{R}^n$ such that
   $$a \in V_a \subset \Omega.$$
   Define
   $$V := \lambda V_a + (1 - \lambda)b = \{\, \lambda x + (1 - \lambda)b \mid x \in V_a \,\}.$$
   Then V is open, contains $\lambda a + (1 - \lambda)b$, and satisfies $V \subset \Omega$ by the convexity of Ω. Hence $\lambda a + (1 - \lambda)b \in \text{int}(\Omega)$, proving that $\text{int}(\Omega)$ is convex.

   Convexity of $\overline{\Omega}$**:** Fix $a, b \in \overline{\Omega}$ and $\lambda \in (0, 1)$. Then there exist sequences $\{a_k\}, \{b_k\} \subset \Omega$ such that $a_k \to a$ and $b_k \to b$. By the convexity of Ω,
   $$\lambda a_k + (1 - \lambda)b_k \in \Omega \quad \text{for all } k.$$
   Taking limits and using continuity of addition and scalar multiplication, we obtain
   $$\lambda a_k + (1 - \lambda)b_k \to \lambda a + (1 - \lambda)b.$$
   Since $\overline{\Omega}$ is closed, it follows that $\lambda a + (1 - \lambda)b \in \overline{\Omega}$. Thus $\overline{\Omega}$ is convex. $\square$

   Remark. Formally, one should consider two cases. If $\text{int}(\Omega) = \varnothing$, then $\text{int}(\Omega)$ is convex trivially. If $\text{int}(\Omega) \neq \varnothing$, the above argument proves its convexity.

   Similarly, if $\overline{\Omega} = \varnothing$ there is nothing to prove. If $\overline{\Omega} \neq \varnothing$, its convexity follows by approximating points in the closure using sequences from Ω.
   These points were discussed in the tutorial and hence omitted here for brevity; however, in an exam setting, such case distinctions should be stated explicitly.

2. Let $f_i : \mathbb{R}^n \to \mathbb{R}$ be convex functions for $i = 1, \ldots, m$.

   (a) Prove that the following functions are convex:

      (i) The scalar multiple αf_i for any $\alpha > 0$.

      (ii) The sum function $\displaystyle\sum_{i=1}^{m} f_i$.

      (iii) The maximum function $\displaystyle\max_{1 \le i \le m} f_i$.

Proof. **(i) Scalar multiple.** Let $\alpha > 0$ and fix any $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$. Since f_i is convex,

$$f_i(\lambda x + (1 - \lambda)y) \le \lambda f_i(x) + (1 - \lambda) f_i(y).$$

Multiplying both sides by α shows that αf_i is convex.

(ii) Sum of convex functions. Define

$$f(x) := \sum_{i=1}^{m} f_i(x).$$

Fix $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$. Using convexity of each f_i, we have

$$f_i(\lambda x + (1 - \lambda)y) \le \lambda f_i(x) + (1 - \lambda) f_i(y), \quad i = 1, \ldots, m.$$

Summing over $i = 1$ to m yields

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= \sum_{i=1}^{m} f_i(\lambda x + (1 - \lambda)y) \\ &\le \lambda \sum_{i=1}^{m} f_i(x) + (1 - \lambda) \sum_{i=1}^{m} f_i(y) \\ &= \lambda f(x) + (1 - \lambda) f(y), \end{aligned}$$

which proves that $\sum_{i=1}^{m} f_i$ is convex.

(iii) Maximum of convex functions.
Recall that the epigraph of a function $h : \mathbb{R}^n \to \mathbb{R}$ is defined as

$$\operatorname{epi}(h) = \{\, (x, y) \in \mathbb{R}^n \times \mathbb{R} \mid h(x) \le y \,\}.$$

Let

$$f(x) := \max_{1 \le i \le m} f_i(x).$$

Then, by definition,

$$(x, y) \in \operatorname{epi}(f) \iff \max_{1 \le i \le m} f_i(x) \le y \iff f_i(x) \le y \quad \text{for all } i = 1, \ldots, m.$$

Hence,

$$\operatorname{epi}(f) = \bigcap_{i=1}^{m} \{\, (x, y) \mid f_i(x) \le y \,\} = \bigcap_{i=1}^{m} \operatorname{epi}(f_i).$$

Since each f_i is convex, $\operatorname{epi}(f_i)$ is a convex set. The intersection of convex sets is convex; hence $\operatorname{epi}(f)$ is convex. Therefore, f is a convex function. $\square$

(b) Let $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ be convex functions. Is the function

$$g(x) := \min\{f_1(x), f_2(x)\}$$

necessarily convex? Justify your answer.

Solution. The answer is *no*; the function $g(x) = \min\{f_1(x), f_2(x)\}$ is not necessarily convex.

Consider the convex functions

$$f_1(x) = x, \qquad f_2(x) = x^2,$$

defined on $\mathbb{R}$. Define

$$g(x) := \min\{x, x^2\}.$$

Choose $x_1 = \frac{1}{2}$, $x_2 = 2$, and $\lambda = \frac{1}{2}$. Then

$$g(x_1) = \min\left\{\tfrac{1}{2}, \tfrac{1}{4}\right\} = \tfrac{1}{4}, \qquad g(x_2) = \min\{2, 4\} = 2.$$

However,

$$g\left(\tfrac{x_1+x_2}{2}\right) = g\left(\tfrac{5}{4}\right) = \min\left\{\tfrac{5}{4}, \tfrac{25}{16}\right\} = \tfrac{5}{4},$$

while

$$\tfrac{1}{2}g(x_1) + \tfrac{1}{2}g(x_2) = \tfrac{1}{2} \cdot \tfrac{1}{4} + \tfrac{1}{2} \cdot 2 = \tfrac{9}{8}.$$

Since

$$g\left(\tfrac{x_1+x_2}{2}\right) > \tfrac{1}{2}g(x_1) + \tfrac{1}{2}g(x_2),$$

the convexity inequality is violated. Hence, g is not convex. $\square$

Remark. (a)(iii) The above argument can be generalized to an arbitrary collection of convex functions. However, since we are not working in the extended real-valued setting, this case has been omitted, as the pointwise supremum may take the value $+\infty$.

One way to avoid this issue is to assume that the family $\{f_i(x)\}_{i \in \mathcal{I}}$ is bounded above for every x. Under this assumption, the exact same epigraph-based argument can be mimicked without modification.

3. **Convexity under Partial Minimization:** We have seen that the maximum of a finite family of convex functions is convex. In a similar spirit, one can show that the pointwise supremum of an arbitrary family of convex functions is also convex. In particular, the function

$$g(x) := \sup_{i \in \mathcal{I}} f_i(x),$$

where each $f_i : \mathbb{R}^n \to \mathbb{R}$ is convex for all $i \in \mathcal{I}$, is convex.

It turns out that certain special forms of minimization also preserve convexity.

Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ be a convex function in the variables (x, y), and let $C \subset \mathbb{R}^m$ be a nonempty convex set. Define

$$g(x) := \inf_{y \in C} f(x, y).$$

Assume that $g(x) > -\infty$ for all $x \in \mathbb{R}^n$. Prove that g is a convex function in x.

Proof. Let $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in (0, 1)$. We must show that

$$g(\lambda x_1 + (1-\lambda)x_2) \leq \lambda g(x_1) + (1-\lambda)g(x_2).$$

Fix $\varepsilon > 0$. By definition of g, there exist points $y_1, y_2 \in C$ such that

$$f(x_1, y_1) \leq g(x_1) + \varepsilon, \qquad f(x_2, y_2) \leq g(x_2) + \varepsilon.$$

Since C is convex, the point

$$y_\lambda := \lambda y_1 + (1-\lambda)y_2 \in C.$$

By definition of g, we have

$$g(\lambda x_1 + (1-\lambda)x_2) = \inf_{y \in C} f(\lambda x_1 + (1-\lambda)x_2, y) \leq f(\lambda x_1 + (1-\lambda)x_2, y_\lambda).$$

Using the convexity of f, it follows that

$$\begin{aligned} g(\lambda x_1 + (1-\lambda)x_2) &\leq f(\lambda x_1 + (1-\lambda)x_2, y_\lambda) \\ &\leq \lambda f(x_1, y_1) + (1-\lambda) f(x_2, y_2) \\ &\leq \lambda g(x_1) + (1-\lambda)g(x_2) + \varepsilon. \end{aligned}$$

$$g(\lambda x_1 + (1-\lambda)x_2) \leq \lambda g(x_1) + (1-\lambda)g(x_2) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \downarrow 0$ gives

$$g(\lambda x_1 + (1-\lambda)x_2) \leq \lambda g(x_1) + (1-\lambda)g(x_2).$$

Therefore, g is a convex function on $\mathbb{R}^n$. $\square$

4. **Infimal Convolution:** Let $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ be convex functions. Define their infimal convolution by

$$(f_1 \star f_2)(x) := \inf \{ f_1(x_1) + f_2(x_2) \mid x_1 + x_2 = x \}.$$

Assume that $(f_1 \star f_2)(x) > -\infty$ for all $x \in \mathbb{R}^n$. Prove that the function $f_1 \star f_2$ is convex on $\mathbb{R}^n$. **Hint: You may use the result proved in Question 3.**

Proof. Define the function

$$f(x, x_1) := f_1(x_1) + f_2(x - x_1), \qquad (x, x_1) \in \mathbb{R}^n \times \mathbb{R}^n.$$

Since f_1 and f_2 are convex, and the map $(x, x_1) \mapsto x - x_1$ is affine, it follows that f is convex in the variables (x, x_1).

By definition of the infimal convolution,

$$(f_1 \star f_2)(x) = \inf_{x_1 \in \mathbb{R}^n} f(x, x_1).$$

Since $(f_1 \star f_2)(x) > -\infty$ for all x, we may apply the result proved in Question 3 with $C = \mathbb{R}^n$. Hence, the function $f_1 \star f_2$ is convex on $\mathbb{R}^n$. $\square$

5. **Distance Function:** Let $\Omega \subset \mathbb{R}^n$ be a nonempty set. Define the distance function associated with Ω by

$$d(x, \Omega) := \inf \{\|x - \omega\| \mid \omega \in \Omega\}.$$

Recall that a mapping $f : \mathbb{R}^n \to \mathbb{R}^p$ is said to be Lipschitz continuous with constant $L > 0$ on a set $C \subset \mathbb{R}^n$ if

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in C.$$

Answer the following:

(i) Prove that $d(x, \Omega) = 0$ if and only if $x \in \overline{\Omega}$.

(ii) Show that the distance function $d(\cdot, \Omega)$ is Lipschitz continuous on $\mathbb{R}^n$ with Lipschitz constant $L = 1$.

Proof. **(i)** Suppose that $d(x, \Omega) = 0$. For each $k \in \mathbb{N}$, by definition of the infimum, there exists $\omega_k \in \Omega$ such that

$$\| x - \omega_k \| < d(x, \Omega) + \frac{1}{k} = \frac{1}{k}.$$

Hence $\omega_k \to x$ as $k \to \infty$, which implies $x \in \overline{\Omega}$.

Conversely, let $x \in \overline{\Omega}$. Then there exists a sequence $\{\omega_k\} \subset \Omega$ such that $\omega_k \to x$. For every $k \in \mathbb{N}$,

$$0 \leq d(x, \Omega) \leq \|x - \omega_k\|.$$

Letting $k \to \infty$ yields $d(x, \Omega) = 0$.

(ii) Fix any $\omega \in \Omega$. By the triangle inequality,

$$\|x - \omega\| \leq \|x - y\| + \|y - \omega\|.$$

Taking the infimum over all $\omega \in \Omega$ on the right-hand side gives

$$d(x, \Omega) \leq \|x - y\| + d(y, \Omega).$$

Interchanging the roles of x and y, we similarly obtain

$$d(y, \Omega) \leq \|x - y\| + d(x, \Omega).$$

Combining the two inequalities yields

$$|d(x, \Omega) - d(y, \Omega)| \leq \|x - y\|.$$

Therefore, the distance function $d(\cdot, \Omega)$ is Lipschitz continuous on $\mathbb{R}^n$ with Lipschitz constant $L = 1$. $\square$

6. Show that the geometric mean

$$f(x) = \left(\prod_{i=1}^{n} x_i \right)^{1/n}$$

is a concave function on its domain

$$\operatorname{dom} f = \mathbb{R}^n_{++}.$$

Proof. Consider the geometric mean

$$f(x) = \left(\prod_{i=1}^{n} x_i \right)^{1/n}, \qquad \operatorname{dom} f = \mathbb{R}^n_{++}.$$

The Hessian matrix $\nabla^2 f(x)$ has entries

$$\frac{\partial^2 f(x)}{\partial x_k^2} = -\frac{n-1}{n^2} \frac{\left(\prod_{i=1}^n x_i\right)^{1/n}}{x_k^2}, \qquad \frac{\partial^2 f(x)}{\partial x_k \partial x_\ell} = \frac{1}{n^2} \frac{\left(\prod_{i=1}^n x_i\right)^{1/n}}{x_k x_\ell}, \quad k \neq \ell.$$

Hence, the Hessian can be written as

$$\nabla^2 f(x) = -\frac{\left(\prod_{i=1}^n x_i\right)^{1/n}}{n^2} \left(n \operatorname{diag}\left(\frac{1}{x_1^2}, \ldots, \frac{1}{x_n^2} \right) - qq^\top \right),$$

where $q \in \mathbb{R}^n$ is defined by $q_i = \frac{1}{x_i}$.

To establish concavity, it suffices to show that $\nabla^2 f(x) \preceq 0$. For any $v \in \mathbb{R}^n$, we compute

$$v^\top \nabla^2 f(x) v = -\frac{\left(\prod_{i=1}^n x_i\right)^{1/n}}{n^2} \left(n \sum_{i=1}^n \frac{v_i^2}{x_i^2} - \left(\sum_{i=1}^n \frac{v_i}{x_i} \right)^2 \right).$$

By the Cauchy–Schwarz inequality applied to the vectors $a = (1, \ldots, 1)$ and $b = (v_1/x_1, \ldots, v_n/x_n)$,

$$\left(\sum_{i=1}^n \frac{v_i}{x_i} \right)^2 \leq n \sum_{i=1}^n \frac{v_i^2}{x_i^2}.$$

Therefore,

$$v^\top \nabla^2 f(x) v \leq 0 \quad \text{for all } v \in \mathbb{R}^n,$$

which shows that $\nabla^2 f(x)$ is negative semidefinite. Hence, f is concave on $\mathbb{R}^n_{++}$. $\square$

7. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. We say that f is *strongly convex with modulus* $c > 0$ if the function

$$x \mapsto f(x) - \frac{c}{2}\|x\|^2$$

is convex.

Prove that the following statements are equivalent for any C^1 function $f : \mathbb{R}^n \to \mathbb{R}$:

(a) f is strongly convex with modulus c.

(b) For all $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$,

$$f(\lambda y + (1-\lambda)x) \leq \lambda f(y) + (1-\lambda)f(x) - \frac{c}{2}\lambda(1-\lambda)\|y - x\|^2.$$

(c) For all $x, y \in \mathbb{R}^n$,

$$f(y) \geq f(x) + \langle \nabla f(x),\, y - x\rangle + \frac{c}{2}\|y - x\|^2.$$

Proof. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a C^1 function and let $c > 0$.

(1) $\Rightarrow$ (2). Assume that f is strongly convex with modulus c, i.e., the function

$$h(x) := f(x) - \frac{c}{2}\|x\|^2$$

is convex. Then, for all $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$,

$$h(\lambda y + (1-\lambda)x) \leq \lambda h(y) + (1-\lambda)h(x).$$

Substituting the definition of h and rearranging terms gives

$$f(\lambda y + (1-\lambda)x) \leq \lambda f(y) + (1-\lambda)f(x) - \frac{c}{2}\lambda(1-\lambda)\|y - x\|^2,$$

which proves (2).

(2) $\Rightarrow$ (3). Fix $x, y \in \mathbb{R}^n$ and define

$$\phi(\lambda) = f\big(x + \lambda(y - x)\big), \qquad \lambda \in [0, 1].$$

By assumption (2),

$$\phi(\lambda) \leq (1-\lambda)\phi(0) + \lambda\phi(1) - \frac{c}{2}\lambda(1-\lambda)\|y - x\|^2.$$

Subtracting $\phi(0)$, dividing by λ, and letting $\lambda \downarrow 0$, we obtain

$$\phi'(0) \leq \phi(1) - \phi(0) - \frac{c}{2}\|y - x\|^2.$$

Since $\phi'(0) = \langle \nabla f(x), y - x\rangle$, this yields

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle + \frac{c}{2}\|y - x\|^2,$$

which proves (3).

(3) $\Rightarrow$ (1). Assume that (3) holds. Define again

$$h(x) := f(x) - \frac{c}{2}\|x\|^2.$$

Then h is differentiable and

$$\nabla h(x) = \nabla f(x) - cx.$$

Using (3), for all $x, y \in \mathbb{R}^n$,

$$\begin{aligned} h(y) &= f(y) - \frac{c}{2}\|y\|^2 \\ &\geq f(x) + \langle \nabla f(x), y - x\rangle + \frac{c}{2}\|y - x\|^2 - \frac{c}{2}\|y\|^2 \\ &= h(x) + \langle \nabla h(x), y - x\rangle. \end{aligned}$$

Thus, h satisfies the first-order condition for convexity, and hence h is convex. Therefore, f is strongly convex with modulus c.

This completes the proof of the equivalence. $\square$

Remark. For twice continuously differentiable functions, there is an equivalent second-order characterization of strong convexity:

$$f \text{ is strongly convex with modulus } c \quad \Longleftrightarrow \quad \nabla^2 f(x) \succeq cI \ \text{ for all } x.$$

This characterization has been omitted here to keep the tutorial concise.
Students are encouraged to prove this result independently for a better understanding of strong convexity.

8. Let $f : \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function. Show that the gradient of f is L-Lipschitz continuous if and only if

$$|f''(x)| \leq L \quad \text{for all } x \in \mathbb{R}.$$

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable.

($\Rightarrow$) Assume that the gradient f' is L-Lipschitz continuous, i.e.,

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in \mathbb{R}.$$

Dividing both sides by $|x - y|$ (for $x \neq y$) gives

$$\frac{|f'(x) - f'(y)|}{|x - y|} \leq L.$$

Letting $y \to x$ and using the differentiability of f', we obtain

$$|f''(x)| \leq L \quad \text{for all } x \in \mathbb{R}.$$

($\Leftarrow$) Conversely, assume that

$$|f''(x)| \leq L \quad \text{for all } x \in \mathbb{R}.$$

Fix any $x, y \in \mathbb{R}$. By the mean value theorem applied to f', there exists ξ between x and y such that

$$f'(x) - f'(y) = f''(\xi)(x - y).$$

Taking absolute values and using the bound on f'' yields

$$|f'(x) - f'(y)| \leq L|x - y|.$$

Thus, f' is L-Lipschitz continuous.

Combining both directions proves the equivalence. $\square$

9. Check whether the following functions are strongly convex and whether their gradients are Lipschitz continuous:

$$(a)\ f(x) = \|x\|_2^2, \qquad (b)\ f(x) = e^x, \qquad (c)\ f(x) = x\log x + x^2 \ \ (x > 0).$$

Solution. We analyze each function separately.

(a) $f(x) = \|x\|_2^2$ **on** $\mathbb{R}^n$**.**

Consider

$$f(x) - \frac{c}{2}\|x\|^2 = \left(1 - \frac{c}{2}\right)\|x\|^2.$$

This function is convex if and only if $1 - \frac{c}{2} \geq 0$, i.e., $c \leq 2$. Hence, f is strongly convex with modulus $c = 2$.

Moreover,

$$\nabla f(x) = 2x, \qquad \|\nabla f(x) - \nabla f(y)\| = 2\|x - y\|,$$

so ∇f is Lipschitz continuous with constant $L = 2$.

(b) $f(x) = e^x$ **on** $\mathbb{R}$**.**

For any $c > 0$, consider

$$f(x) - \frac{c}{2}x^2 = e^x - \frac{c}{2}x^2.$$

This function is not convex on $\mathbb{R}$, since the quadratic term dominates for large negative x. Hence, f is not strongly convex on $\mathbb{R}$.

Further, since

$$f'(x) = e^x, \qquad f''(x) = e^x,$$

and $f''(x)$ is unbounded on $\mathbb{R}$, the gradient ∇f is not Lipschitz continuous.

(c) $f(x) = x\log x + x^2$**,** $\quad x > 0$**.**

Consider

$$f(x) - \frac{c}{2}x^2 = x\log x + \left(1 - \frac{c}{2}\right)x^2.$$

The function $x\log x$ is convex on $(0, \infty)$, and x^2 is convex on $\mathbb{R}$. Thus, the above function is convex provided $1 - \frac{c}{2} \geq 0$, i.e., $c \leq 2$. Hence, f is strongly convex on $(0, \infty)$ with modulus $c = 2$.

Moreover,

$$f'(x) = \log x + 1 + 2x, \qquad f''(x) = \frac{1}{x} + 2.$$

Since $f''(x)$ is unbounded as $x \downarrow 0$, the gradient ∇f is not Lipschitz continuous on $(0, \infty)$.

This completes the solution. $\square$

10. **Robust Optimization:** In some optimization problems, there is uncertainty or variation in the objective and constraint functions due to parameters or factors that are either unknown or beyond our control. We can model this situation by allowing the objective and constraint functions $f_0, \ldots, f_m$ to depend on the optimization variable $x \in \mathbb{R}^n$ and a parameter vector $u \in \mathbb{R}^k$ that is unknown or varying.

    In the stochastic optimization formulation, we work with the expected values $\mathbb{E}_u\, f_i(x, u)$. In the worst-case analysis approach, we are given a set $\mathcal{U}$ in which u is known to lie, and we work with the maximum or worst-case values $\sup_{u \in \mathcal{U}} f_i(x, u)$. To simplify the discussion, we assume that there are no equality constraints.

    (a) **Stochastic optimization:** Consider the optimization problem

    $$\begin{array}{ll} \text{minimize} & \mathbb{E}\, f_0(x, u) \\ \text{subject to} & \mathbb{E}\, f_i(x, u) \leq 0, \quad i = 1, \ldots, m, \end{array}$$

    where the expectation is taken with respect to the random parameter u. Show that if each function $f_i(x, u)$ is convex in x for every fixed u, then the above stochastic optimization problem is convex.

    (b) **Worst-case optimization:** Consider the optimization problem

    $$\begin{array}{ll} \text{minimize} & \sup\limits_{u \in \mathcal{U}} f_0(x, u) \\ \text{subject to} & \sup\limits_{u \in \mathcal{U}} f_i(x, u) \leq 0, \quad i = 1, \ldots, m, \end{array}$$

    where $\mathcal{U} \subset \mathbb{R}^k$ is a given uncertainty set. Show that if each function $f_i(x, u)$ is convex in x for every fixed u, then this worst-case optimization problem is convex.

Proof. **(a) Stochastic optimization.** Assume that for each fixed u, the function $f_i(x, u)$ is convex in x. Then, for all $x, y \in \mathbb{R}^n$ and $\theta \in (0, 1)$,

$$f_i(\theta x + (1 - \theta)y, u) \leq \theta f_i(x, u) + (1 - \theta) f_i(y, u) \quad \text{a.s.}$$

Taking expectations on both sides and using the monotonicity and linearity of expectation, we obtain

$$\mathbb{E}\, f_i(\theta x + (1 - \theta)y, u) \leq \theta \mathbb{E}\, f_i(x, u) + (1 - \theta)\mathbb{E}\, f_i(y, u).$$

Thus, $\mathbb{E}\, f_i(x, u)$ is convex in x. Since the objective function and all constraint functions are convex, the stochastic optimization problem is convex.

(b) Worst-case optimization. Assume that for each fixed $u \in \mathcal{U}$, the function $f_i(x, u)$ is convex in x. Then the pointwise supremum

$$x \mapsto \sup_{u \in \mathcal{U}} f_i(x, u)$$

is convex in x. Since the objective function and all constraint functions in the worst-case formulation are convex, the worst-case optimization problem is convex. $\square$

11. Let $(\mathbb{R}^n, \|\cdot\|)$ be a normed linear space, and let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by

$$f(x) = \|x\|.$$

Recall that $(\mathbb{R}^n)^* = \mathbb{R}^n$(for the following case) and that the *dual norm* $\|\cdot\|_*$ is defined by

$$\|y\|_* \equiv \max_{\|x\|\le 1} y^\top x, \qquad y \in \mathbb{R}^n.$$

Show that the subdifferential of f at $x = 0$ is given by

$$\partial f(0) = \{\, y \in \mathbb{R}^n : \|y\|_* \le 1 \,\}.$$

Hint: You may use the generalized Cauchy–Schwarz inequality

$$|y^\top x| \le \|y\|_* \, \|x\|.$$

Proof. Let $f(x) = \|x\|$ and recall that $y \in \partial f(0)$ if and only if

$$f(x) \ge f(0) + y^\top x \quad \text{for all } x \in \mathbb{R}^n.$$

Since $f(0) = 0$, this condition becomes

$$\|x\| \ge y^\top x \quad \text{for all } x \in \mathbb{R}^n.$$

($\subseteq$) Assume $y \in \partial f(0)$. Then $\|x\| \ge y^\top x$ for all x. Taking the maximum over all x with $\|x\| \le 1$, we obtain

$$\max_{\|x\|\le 1} y^\top x \le 1,$$

which implies $\|y\|_* \le 1$.

($\supseteq$) Conversely, assume $\|y\|_* \le 1$. By the generalized Cauchy–Schwarz inequality,

$$y^\top x \le \|y\|_* \, \|x\| \le \|x\| \quad \text{for all } x \in \mathbb{R}^n.$$

Hence, $\|x\| \ge y^\top x$ for all x, which shows that $y \in \partial f(0)$.

Therefore,

$$\partial f(0) = \{\, y \in \mathbb{R}^n : \|y\|_* \le 1 \,\}.$$

□

Remark. Although the generalized Cauchy–Schwarz inequality is stated as a hint, it is easy to verify directly. In the definition of the dual norm, one may replace x by $x/\|x\|$; since the maximum is taken over all vectors with norm at most one, this immediately yields the desired inequality. The reader is encouraged to prove the reverse inequality independently.
As a useful fact from linear algebra, the dual norm of the ℓ_1 norm is the ℓ_∞ norm, and the dual norm of the ℓ_2 norm is the ℓ_2 norm.

12. Give an example of two closed convex sets in $\mathbb{R}^n$ that are disjoint but cannot be strictly separated by a hyperplane.

Solution. Let

$$A := \operatorname{epi}(e^x) = \{(x, y) \in \mathbb{R}^2 \mid y \geq e^x\}, \qquad B := \{(x, y) \in \mathbb{R}^2 \mid y \leq 0\}.$$

Both A and B are closed and convex, and they are disjoint since $e^x > 0$. Moreover, $e^x \to 0$ as $x \to -\infty$, so the distance between A and B is zero and no strict separating hyperplane exists. $\square$

13. **Inner and outer polyhedral approximations:** Let $C \subseteq \mathbb{R}^n$ be a closed convex set, and let $x_1, \ldots, x_K$ be points on the boundary of C. Suppose that for each $i = 1, \ldots, K$, the hyperplane

$$a_i^\top (x - x_i) = 0$$

is a supporting hyperplane of C at x_i, i.e.,

$$C \subseteq \{\, x \mid a_i^\top (x - x_i) \leq 0 \,\}.$$

Define the polyhedra

$$P_{\text{inner}} = \operatorname{conv}\{x_1, \ldots, x_K\}, \qquad P_{\text{outer}} = \{\, x \mid a_i^\top (x - x_i) \leq 0,\ i = 1, \ldots, K \,\}.$$

Show that

$$P_{\text{inner}} \subseteq C \subseteq P_{\text{outer}}.$$

Illustrate this result with a suitable diagram.

Proof. Since $x_1, \ldots, x_K$ lie on the boundary of C and C is closed, we have $x_i \in C$ for all $i = 1, \ldots, K$. Because C is convex, any convex combination of $x_1, \ldots, x_K$ also belongs to C. Hence,

$$P_{\text{inner}} = \operatorname{conv}\{x_1, \ldots, x_K\} \subseteq C.$$

Now let $x \in C$. Since each hyperplane $a_i^\top (x - x_i) = 0$ is a supporting hyperplane of C at x_i, we have

$$a_i^\top (x - x_i) \leq 0 \quad \text{for } i = 1, \ldots, K.$$

Therefore, $x \in P_{\text{outer}}$, and hence $C \subseteq P_{\text{outer}}$. $\square$

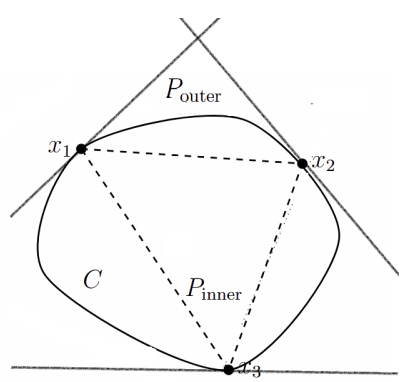

Illustration of the inclusions $P_{\text{inner}} \subseteq C \subseteq P_{\text{outer}}$.

14. Recall from class that you have studied the subgradient descent method and the proximal gradient descent method for minimizing convex functions.

Subgradient Descent	**Proximal Gradient Descent**
f may be non-smooth	f is differentiable, g may be non-smooth
$\min_x f(x)$	$\min_x \; f(x) + g(x)$
$x^{k+1} = x^k - \alpha_k g^k, \quad g^k \in \partial f(x^k),\ \alpha_k > 0$	$x^{k+1} = \mathrm{prox}_{\alpha_k g}\big(x^k - \alpha_k \nabla f(x^k)\big), \quad \alpha_k > 0$
	where $\mathrm{prox}_{\alpha g}(z) = \arg\min_x \left\{ g(x) + \frac{1}{2\alpha}\|x - z\|^2 \right\}$

Now consider the one-dimensional function

$$F(x) = \frac{1}{2}x^2 + |x|.$$

Write the explicit update rules for both subgradient descent and proximal gradient descent when applied to the function F.

Solution. Consider the function

$$F(x) = \frac{1}{2}x^2 + |x| = f(x) + g(x),$$

where

$$f(x) = \frac{1}{2}x^2, \qquad g(x) = |x|.$$

Subgradient descent. A subgradient of F at x is given by

$$\partial F(x) = \begin{cases} \{x+1\}, & x > 0, \\ \{x-1\}, & x < 0, \\ [-1, 1], & x = 0, \end{cases} \qquad \text{(recall from Lecture 10: } \partial(f+g) = \nabla f + \partial g\text{)}.$$

Hence, the subgradient descent update

$$x^{k+1} = x^k - \alpha_k g^k, \qquad g^k \in \partial F(x^k),$$

takes the explicit form

$$\boxed{x^{k+1} = \begin{cases} x^k - \alpha_k(x^k + 1), & x^k > 0, \\ x^k - \alpha_k(x^k - 1), & x^k < 0, \\ x^k - \alpha_k s, \quad s \in [-1, 1], & x^k = 0. \end{cases}}$$

Proximal gradient descent. The gradient of f is $\nabla f(x) = x$. The proximal gradient update is

$$x^{k+1} = \mathrm{prox}_{\alpha_k g}\big(x^k - \alpha_k \nabla f(x^k)\big) = \mathrm{prox}_{\alpha_k |\cdot|}\big((1-\alpha_k)x^k\big)\,.$$

The proximal operator of the absolute value is the soft-thresholding operator(discussed in class)

$$\mathrm{prox}_{\alpha|\cdot|}(z) = \begin{cases} z-\alpha, & z > \alpha, \\ 0, & |z| \le \alpha, \\ z+\alpha, & z < -\alpha. \end{cases}$$

Therefore, the update rule becomes

$$\boxed{x^{k+1} = \begin{cases} (1-\alpha_k)x^k - \alpha_k, & (1-\alpha_k)x^k > \alpha_k, \\ 0, & |(1-\alpha_k)x^k| \le \alpha_k, \\ (1-\alpha_k)x^k + \alpha_k, & (1-\alpha_k)x^k < -\alpha_k. \end{cases}}$$

□

Remark. This is a toy problem intended to illustrate how subgradient descent and proximal gradient descent work in practice. Students are encouraged to run simple numerical simulations to gain further intuition about the behavior of these methods.
The problem naturally generalizes to higher dimensions by replacing $\frac{1}{2}x^2$ with $\frac{1}{2}x^\top Ax$ and $|x|$ with the ℓ_1 norm of x. Students are encouraged to work out this generalized case independently to develop a better understanding of the methods.